

AI Assignment #2

Variational Bayesian Last Layers for
Clothing Image Classification

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Abstract

Thesis: a discriminative variational Bayesian last layer (D-VBLL; Harrison, Willes, and Snoek, ICLR 2024) puts a Gaussian posterior on the last-layer weights of a clothing classifier. On Fashion-MNIST it matches softmax MAP on accuracy, likelihood, and on which errors it ranks as uncertain. Its calibration error is lower than an unscaled softmax, but one temperature fit on the validation set calibrates the softmax further. Gaussian naive Bayes on pixels fails; linear discriminant analysis on the same CNN features is close in accuracy and worse as a probability model. The learned weight variance stays far below the prior, so the posterior predictive stays close to a point estimate, and neither head detects low-contrast shift. Search, implementation, experiments, and this report were done by a Cursor agent under a standing instruction that I would not hand-write code.

1 How I Asked AI Tools to Find the Algorithm

The Moodle brief asks for a *recent* classification or recognition algorithm based on probability or Bayes theory, for computer vision, with the search, code, experiments, and report done by an AI tool. I used Cursor only. I pasted the brief, pointed at my Assignment 1 repository, and gave the agent authority to choose the method. Assignment 1 had already used FasterNet, a CVPR 2023 convolution designed for speed. That model is not a probabilistic classifier, so it could not be reused as the answer to this brief.

The agent searched recent Bayesian deep-learning papers and kept three candidates that are actually about classification uncertainty:

- **Variational Bayesian Last Layers** (Harrison, Willes, and Snoek, ICLR 2024), a sampling-free variational posterior on the last layer.
- **FSP-Laplace** (Cinquin et al., NeurIPS 2024), a Laplace approximation with a function-space Gaussian-process prior.
- **Last Layer Empirical Bayes** (Villemoore et al., 2025), a normalizing-flow prior on the last layer.

The choice was a table, not a preference for the newest title. D-VBLL is the only one of the three whose training loss is a closed form I can check on a toy batch before spending a training run. FSP-Laplace needs a function-space mode and a matrix-free Hessian. Last-layer empirical Bayes adds a normalizing flow on top of the classifier. Fashion-MNIST clothing recognition is also a different task from the digit-robustness study in Assignment 1.

Table 1: Candidates considered for a recent Bayes classifier that a laptop run can verify.

	D-VBLL	FSP-Laplace	LLEB
Venue	ICLR 2024	NeurIPS 2024	2025
Object with a posterior	last-layer W	network function	last-layer flow
Check before training	Theorem 2	weak mode, Hessian	ELBO through a flow
Extra machinery	diagonal Gaussian	function-space GP	normalizing flow
Used in this report	yes	no	no

The diary of prompts, rejected alternatives, and checks is `process_log.md`.

2 Algorithm Description

Model.. A network with weights θ maps an image x to a feature vector $\phi(x, \theta) \in \mathbb{R}^D$. Class logits are a noisy linear readout. For a last-layer matrix W whose k -th row is w_k ,

$$p(y \mid x, W, \theta) = \text{softmax}(z), \quad z = W\phi(x, \theta) + \varepsilon, \quad (1)$$

with ε zero-mean Gaussian noise. This is Bayesian logistic regression on learned features. The prior on each row is isotropic Gaussian, $p(w_k) = \mathcal{N}(0, s^2 I)$, with the authors’ fan-in scaling $s^2 = 2/D$. Exact softmax marginalization is intractable, so training uses a variational posterior that factorizes across classes, $q(w_k) = \mathcal{N}(m_k, \text{diag}(v_k))$. Diagonal covariances drop cross-class correlation and make every loss term linear in D .

Training bound.. Stochastic variational inference [7] maximizes a lower bound on the log marginal likelihood. For one example with label y , Theorem 2 replaces the expected log-softmax by

$$m_y^\top \phi - \log \sum_k \exp\left(m_k^\top \phi + \frac{1}{2}(\phi^\top \text{diag}(v_k)\phi + \sigma_k^2)\right). \quad (2)$$

The extra $\frac{1}{2}\phi^\top \text{diag}(v_k)\phi$ term is the Jensen gap: uncertain weights inflate every logit before the log-sum-exp, which stops the bound from behaving like an over-confident point estimate. The mini-batch loss I minimize is the negative of that bound, plus $(1/T) \text{KL}(q(W) \parallel p(W))$, minus a $(1/T)$ inverse-Wishart-style penalty on the noise variances σ_k^2 . T is the number of training images. Feature weights θ get an ordinary Gaussian prior, implemented as AdamW weight decay. The last-layer parameters are not weight-decayed, because the KL term is already their regularizer. This is the paper’s “full training” objective, not a post-hoc Laplace approximation and not Bayes-by-Backprop on every layer [6].

Prediction.. The posterior predictive probability is the expectation of the softmax under $q(W)$. That expectation has no closed form for the discriminative model, so I draw logit samples

$$z^{(s)} \sim \mathcal{N}(m^\top \phi, \text{diag}(\phi^\top S \phi + \sigma^2)) \quad (3)$$

and average $\text{softmax}(z^{(s)})$. Test metrics use 32 samples. Out-of-distribution scores use the maximum predictive probability [4].

What is Bayesian here.. Write the four objects separately, because a pattern-recognition classifier is the posterior predictive, not the training loss. The prior is $p(W) = \prod_k \mathcal{N}(w_k; 0, s^2 I)$. The likelihood is the softmax in Equation (1). The variational posterior $q(W)$ stands in for $p(W \mid \text{data})$, and the KL term keeps it from ignoring the prior. The reported class probability is the posterior predictive $\mathbb{E}_{q(W)}[\text{softmax}(W\phi)]$, not a point W . A softmax MAP baseline uses the same features and the same likelihood, but one weight matrix. Gaussian naive Bayes and linear discriminant analysis, used later as controls, are the generative version of this story: they put Gaussians on ϕ or on the

pixels and apply Bayes' rule for $p(y | x)$. D-VBLL is the discriminative version. It never builds $p(x | y)$.

Discriminative VBLL classifier used in this assignment

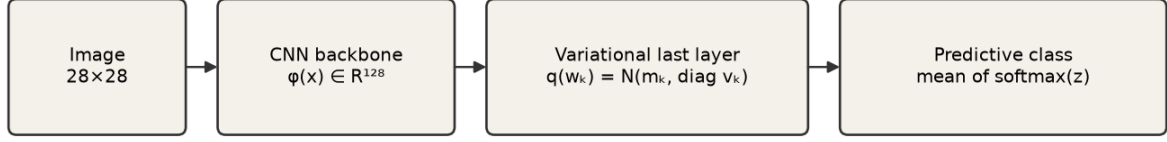


Figure 1: Student-scale D-VBLL. The MAP baseline replaces the variational last layer with one linear layer and a softmax.

3 How AI Implemented the Algorithm

The Cursor agent wrote the repository. I did not hand-write or hand-edit the Python. The layout is:

- `models/vbll.py` — diagonal D-VBLL: Theorem 2 bound, closed-form KL, noise penalty, and Monte Carlo predictive probabilities.
- `models/backbone.py` — shared CNN. Three conv stages, batch-norm, global average pool, 128-d features. About 1.58×10^5 parameters.
- `models/classifier.py` — `map` (cross-entropy) or `vbll` (ELBO). Weight decay applies to the backbone only when the head is Bayesian.
- `data_utils.py` — Fashion-MNIST / MNIST tensors, shared augmentation, pixel-space corruptions.
- `train.py`, `evaluate.py`, `metrics.py`, `figures.py`.
- `run_all.py`, `run_ablation.py`, `extra_analysis.py`. The last script does not retrain.

The training step, in the form that `self_check()` tests, is:

$$\begin{aligned}
 \phi &\leftarrow \text{backbone}(x). \\
 \mu_k &\leftarrow m_k^\top \phi, \quad v_k \leftarrow \phi^\top \text{diag}(v_k) \phi + \sigma_k^2. \\
 \text{bound} &\leftarrow \mu_y - \log \sum_k \exp(\mu_k + \tfrac{1}{2} v_k). \\
 \text{loss} &\leftarrow -\text{mean}(\text{bound}) + (1/T) \text{KL}(q||p) - (1/T) \log p(\sigma^2).
 \end{aligned}$$

Before training, `self_check()` builds an 8×4 random batch and checks three facts: the loss is finite and back-propagates; the bound matches a hand-expanded log-sum-exp; multiplying the features by four increases logit variance. That check is the implementation test. It does not use Fashion-MNIST accuracy as a substitute for the formula.

Practical settings follow the paper and the authors' diagonal parameterization rather than a new variant. The prior variance is $2/D$. Weight standard deviations are initialized near $1/D$. The KL is scaled by $1/T$ in the main runs. I did not import the `vbll` library; the module is a direct implementation of the bound above so the loss can be read against Equation (2).

4 Experiment Settings and Results

Task.. In-distribution data are Fashion-MNIST clothing images, 28×28 greyscale, 10 classes [2]. The official training set is split once, with seed 0, into 54,000 training images and 6,000 validation

images. The 10,000 official test images are used only for the tables below. MNIST digits are a far out-of-distribution set [3]: same resolution, different pattern family. Both datasets are normalized with the Fashion-MNIST training mean 0.286 and standard deviation 0.353.

Protocol.. Both models share the CNN, the augmentation (horizontal flip and a random crop after 2-pixel padding), AdamW at learning rate 10^{-3} , cosine decay, batch size 128, and 12 epochs. Backbone weight decay is 10^{-4} . Runs use two seeds, $\{0, 1\}$, on Apple MPS (PyTorch 2.14). Validation probabilities for D-VBLL use 16 logit samples; the test tables use 32. Reported means are over the two seeds. Calibration is expected calibration error with 15 bins [5].

Table 2: Clean Fashion-MNIST test set. Mean over two seeds.

Model	Accuracy	NLL	ECE	Entropy
Softmax MAP	0.930	0.200	0.0109	0.168
D-VBLL ($1/T$)	0.928	0.200	0.0075	0.183

On clean clothes the two models are tied on accuracy and likelihood. D-VBLL’s ECE is lower on both seeds (MAP 0.0101 and 0.0116; D-VBLL 0.0075 and 0.0076). The absolute gap is small because the MAP model is already well calibrated. Figure 2 shows the shared learning curve. Figure 3 shows that D-VBLL’s reliability curve sits closer to the diagonal through the middle bins.

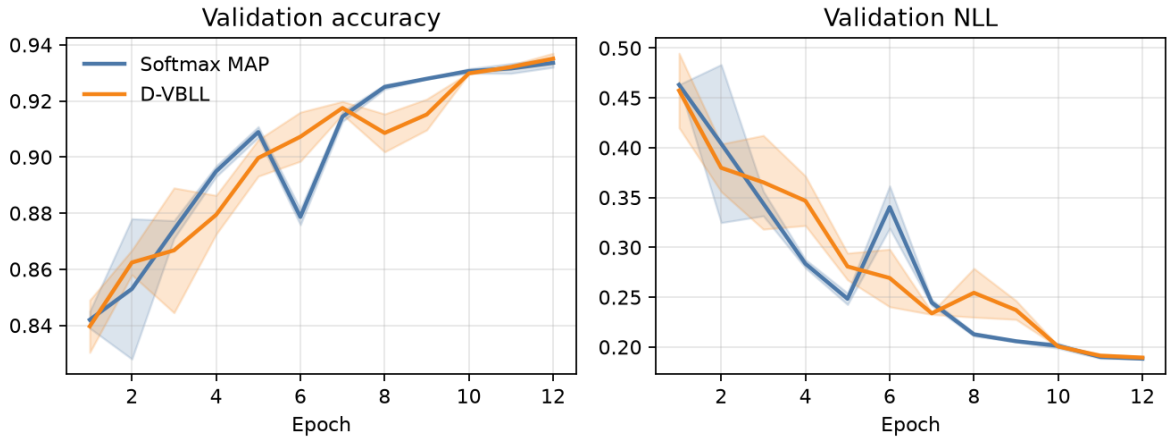


Figure 2: Validation accuracy and NLL. Bands are the two seeds. The training losses are not plotted together because cross-entropy and the ELBO are different objectives.

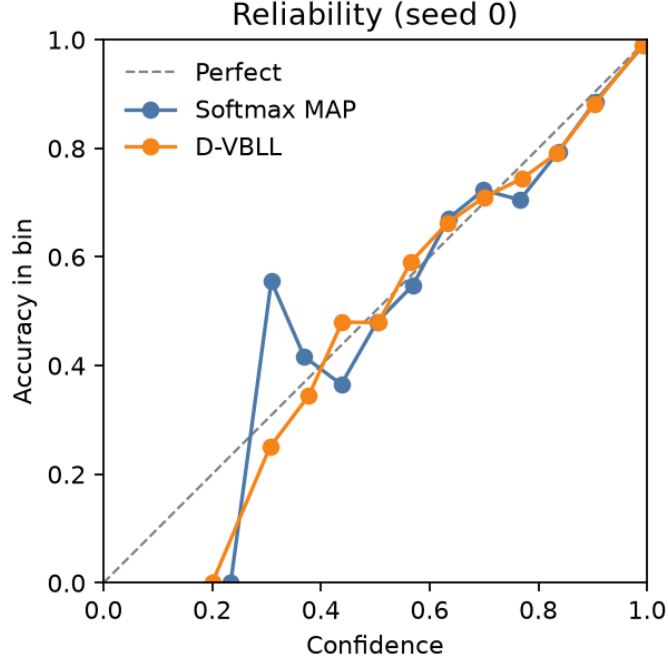


Figure 3: Reliability diagram on the clean test set, seed 0. Both models are close to the diagonal; D-VBLL is the smoother of the two.

Controls a probability model still has to pass.. A lower ECE than softmax is not, by itself, evidence that the posterior did the work. Three controls use the same test set and do not change the trained CNN.

Temperature scaling [5] divides the MAP logits by a scalar T fit on the 6,000 validation images by minimizing NLL. It does not change the predicted class. On the two seeds, T is 1.114 and 1.088. Test ECE falls to 0.0050 and 0.0067 (mean 0.0059), below D-VBLL’s 0.0075, and NLL is 0.198 and 0.199. The ECE gap in Table 2 is therefore mostly a temperature, which matches the later finding that $q(W)$ has concentrated.

Gaussian naive Bayes on raw pixels, with an independent Gaussian per pixel and per class, reaches accuracy 0.579, NLL 7.63, and ECE 0.419. The independence assumption does not describe a garment. Shrinkage linear discriminant analysis on the 128-d features of the seed-0 MAP network is the classical shared-covariance Bayes classifier on the same representation. It reaches accuracy 0.917, close to the discriminative heads, but NLL 0.544 and ECE 0.063. The CNN features carry the class. The softmax and D-VBLL heads are what make the probabilities sharp.

Ranking is a separate question from calibration. Using confidence to detect the model’s own mistakes, AUROC is 0.922 and 0.924 for MAP and 0.923 and 0.926 for D-VBLL. Keeping the most confident 90% of the test set raises accuracy from about 0.930 to 0.968 (MAP) and 0.967 (D-VBLL). Figure 4 shows the risk-coverage curves lying on top of each other. D-VBLL does not rank errors better than softmax once the posterior has collapsed. Both heads are still useful as reject options: the bottom decile of confidence contains a disproportionate share of the mistakes.

Table 3: Clean Fashion-MNIST test. Classical Bayes and temperature scaling are controls, not extra neural nets. Means over two seeds where two runs exist; LDA and pixel naive Bayes are single fits.

Model	Accuracy	NLL	ECE	Error AUROC
Gaussian naive Bayes (pixels)	0.579	7.630	0.419	—
LDA on MAP features	0.917	0.544	0.063	—
Softmax MAP	0.930	0.200	0.0109	0.923
MAP + temperature	0.930	0.199	0.0059	0.923
D-VBLL ($1/T$)	0.928	0.200	0.0075	0.924

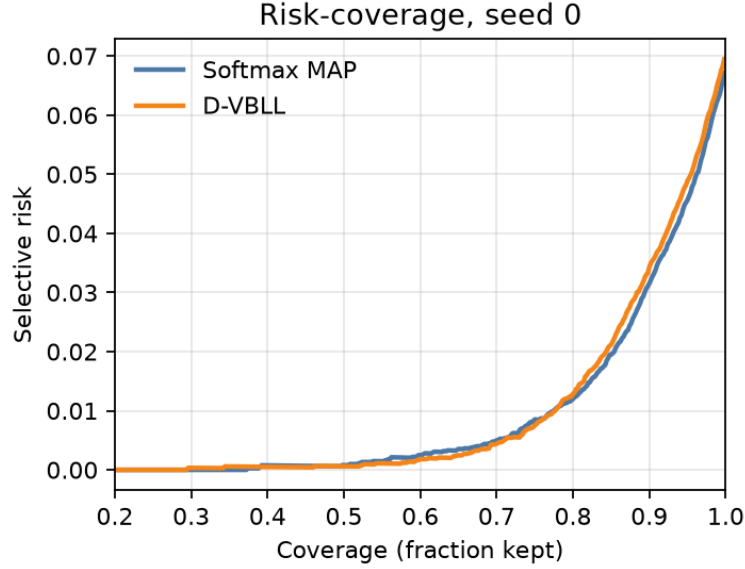


Figure 4: Selective risk against coverage, seed 0. Lower is better. The curves overlap: both heads become accurate once low-confidence predictions are refused.

Where the errors are.. Seed-0 D-VBLL per-class accuracy is high on bags (0.989), trousers (0.987), and sandals (0.978), and lowest on shirts (0.769). The largest off-diagonal counts are shirt→T-shirt (96), T-shirt→shirt (79), shirt→coat (70), and coat→shirt (41). Those pairs share a torso silhouette. Ankle boot→sneaker (40) is the main footwear confusion. Figure 5 is the row-normalized matrix. Figure 6 shows the failure mode the ECE number hides: a few errors still receive predictive probability near 1, including a boot read as a sneaker.

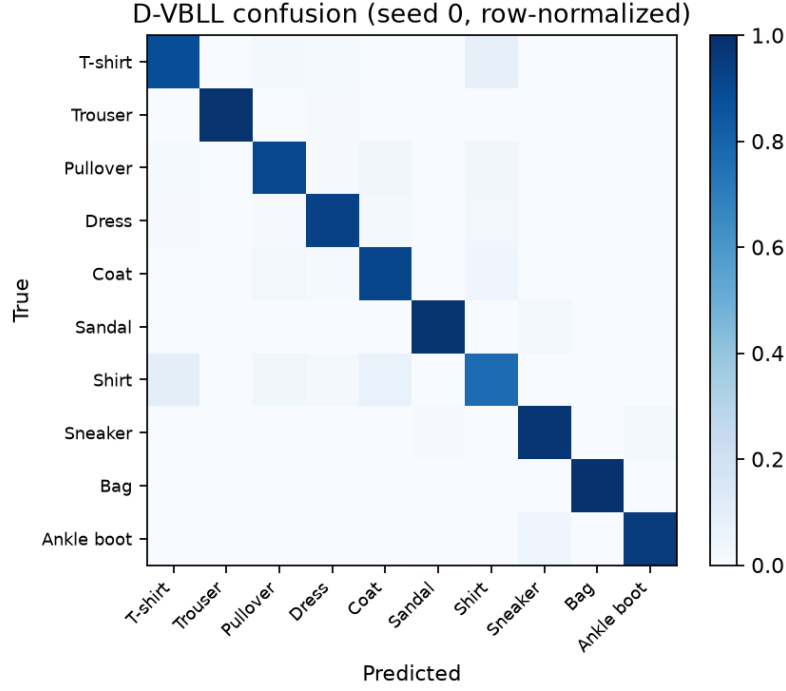


Figure 5: D-VBLL confusion on the clean test set, seed 0. Mass off the diagonal sits in the shirt / T-shirt / coat / pullover block.

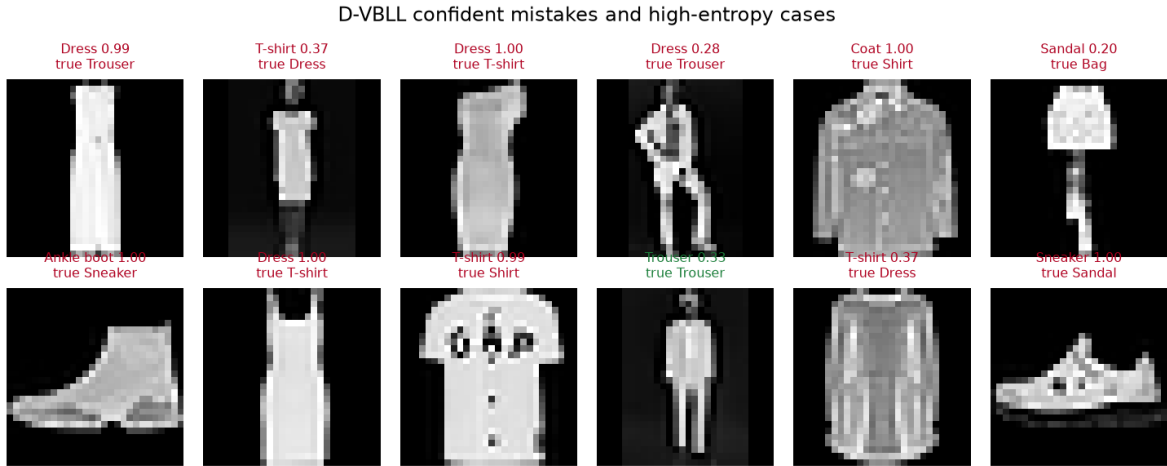
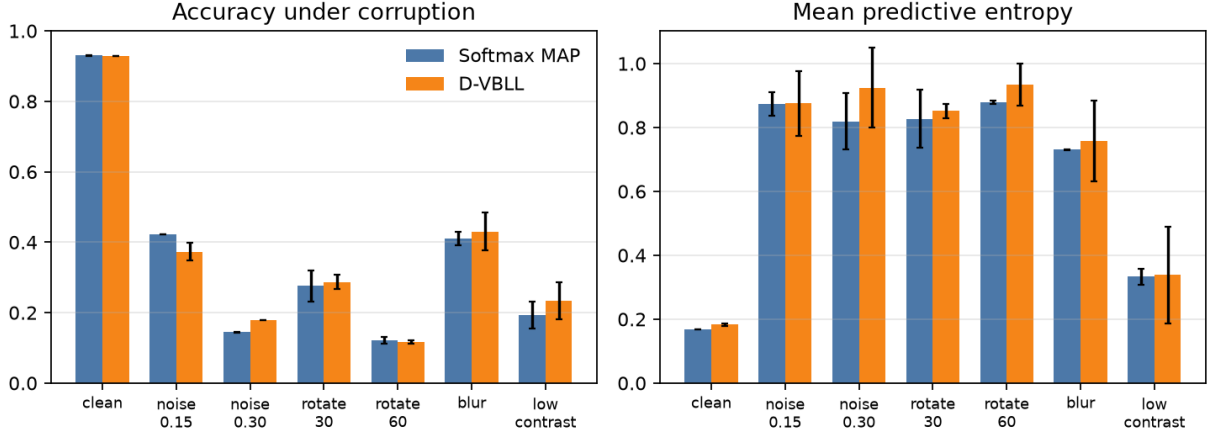


Figure 6: D-VBLL seed 0. Titles are the predictive class, its probability, and the true class. Red titles are mistakes, including high-confidence ones.

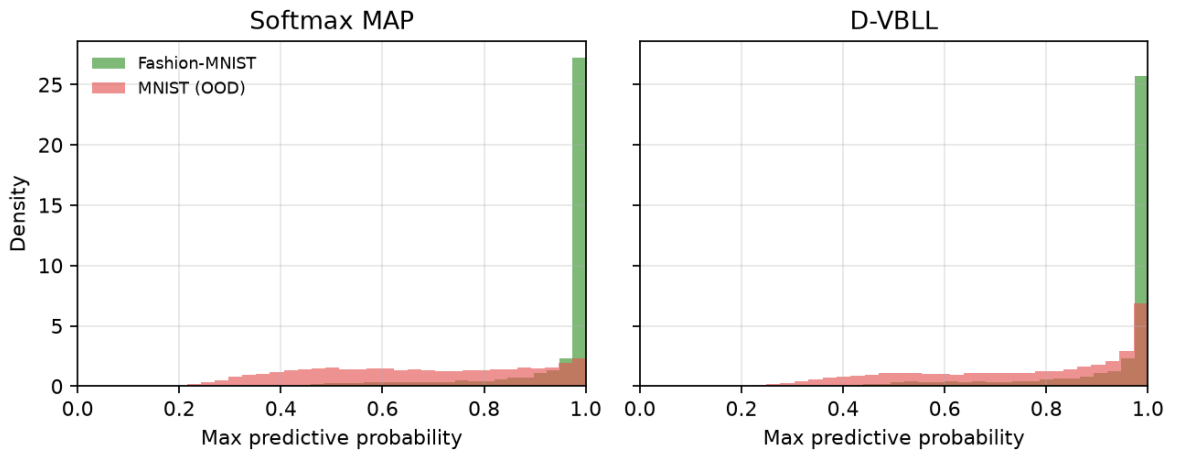
Corruptions.. Corruptions are applied in pixel space at test time only. The models were not trained on them. Table 4 and Figure 7 show that geometric and photometric shift hurts both heads. At noise 0.30, rotation 60° , and low contrast, accuracy falls to roughly 0.12–0.23. Low contrast is the instructive case: accuracy is about 0.19–0.23, but mean entropy stays near 0.33, lower than under noise, and ECE exceeds 0.65. Both models are confidently wrong when the silhouette is preserved and the intensity is flattened. Last-layer weight uncertainty is not a corruption detector.

Table 4: Test accuracy under corruption. Mean over two seeds.

Condition	Softmax MAP	D-VBLL
Clean	0.930	0.928
Gaussian noise 0.15	0.422	0.373
Gaussian noise 0.30	0.145	0.180
Rotate 30°	0.276	0.287
Rotate 60°	0.121	0.117
5 × 5 average blur	0.409	0.430
Low contrast	0.193	0.235

**Figure 7:** Accuracy falls under every corruption. Entropy rises under noise and rotation, but stays low under low contrast, where both models are confidently wrong.

Out-of-distribution digits.. Using maximum predictive probability as the in-distribution score, AUROC against MNIST is 0.902 and 0.808 for the two MAP seeds (mean 0.855), and 0.825 and 0.855 for D-VBLL (mean 0.840). The seed spread is larger than the gap between models, so I do not claim an OOD win for either head. Figure 8 shows why the AUROC is only moderate: most clothes sit near probability 1, and a visible slice of digits does too.

**Figure 8:** Maximum predictive probability, seed 0. MNIST is shifted left of Fashion-MNIST, but both distributions have a spike at 1.

Why the Bayesian head stays close to MAP.. After training at KL weight $1/T$, the mean diagonal variance of W is about 4.9×10^{-4} . The prior variance is $2/D = 0.0156$, roughly thirty times larger, so the posterior has concentrated. The KL itself is about 2.3×10^3 ; divided by $T = 54000$

it contributes only a few hundredths to the loss, which is enough to regularize but not enough to keep the prior’s uncertainty. Figure 9 then has a simple shape for both models: predictive entropy falls as $\|\phi\|$ grows. Large features produce large mean logits, and the quadratic variance term is too small to cancel that peak. This is the opposite of a distance-aware uncertainty method, and it matches the modest ECE gap in Table 2.

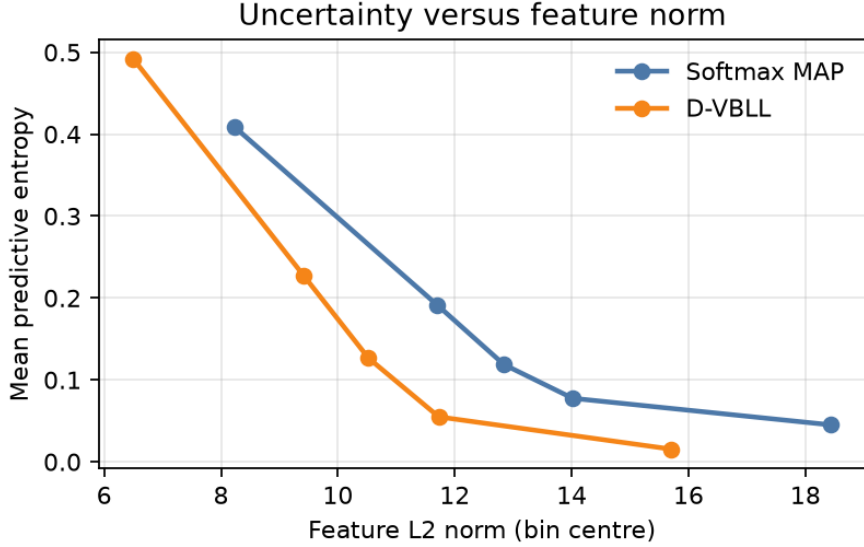


Figure 9: Clean-test entropy against feature norm. Higher-norm features are more confident for both heads.

KL-weight ablation (seed 0).. The $1/T$ factor is a hyperparameter of the ELBO estimator, not a constant of nature. I retrained D-VBLL at $10/T$ and $100/T$ with the same seed and schedule. Table 5 shows the trade. Stronger KL pulls $q(W)$ toward the prior. Mean weight variance goes from 4.9×10^{-4} at $1/T$ to 4.7×10^{-3} at $10/T$ and 8.1×10^{-3} at $100/T$, against a prior variance of 0.0156. The KL value falls from about 2.3×10^3 to 229, and predictive entropy rises from 0.187 to 0.639. Clean accuracy stays at 0.927–0.929 across all three settings. What changes is the shape of the predictive distribution: clean NLL and ECE get worse, because these test images want sharp probabilities, while MNIST AUROC on this seed rises from 0.825 to 0.916. $1/T$, the setting in the main table, is the best calibrated of the three. The ablation shows that the KL term is alive: it trades clean calibration for broader uncertainty, without a collapse in accuracy at this model size.

Table 5: D-VBLL KL multiplier, seed 0, clean test unless noted. Weight variance is the mean diagonal entry of $q(W)$.

KL weight	Accuracy	NLL	ECE	Entropy	MNIST AUROC
$1/T$	0.929	0.201	0.0075	0.187	0.825
$10/T$	0.927	0.212	0.0154	0.280	0.869
$100/T$	0.927	0.289	0.0937	0.639	0.916

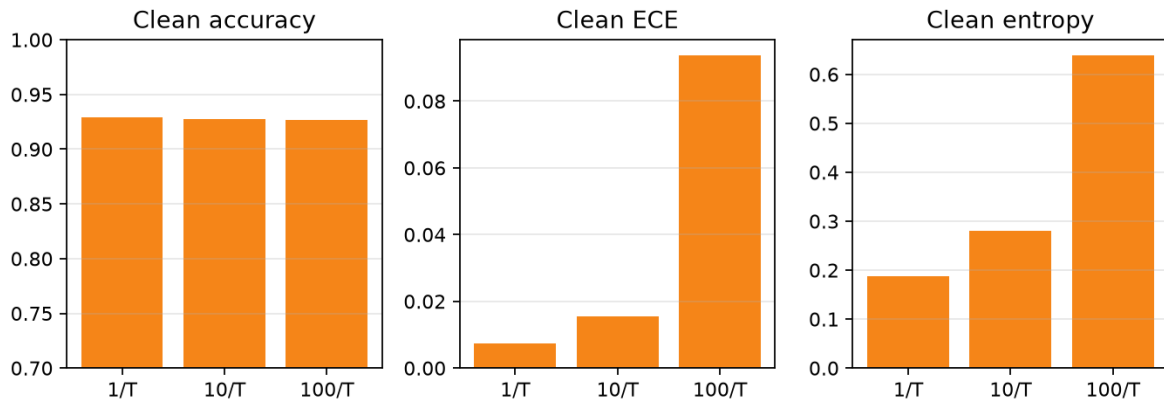


Figure 10: Seed-0 ablation of the KL multiplier. $1/T$ is the main setting.

5 What I Have Learnt from This AI Assignment

- The useful constraint was the brief itself: probability or Bayes, recent, and not a repeat of FasterNet. Once that was stated, the agent could reject a fast CNN and look for a last-layer posterior with a checkable formula.
- A Bayesian classifier is defined by the predictive distribution, not by the word “uncertainty” in the README. Here that distribution is an average of softmaxes under $q(W)$. The self-check on Equation (2) mattered more than the first epoch’s accuracy.
- Matching accuracy is a real result, and it is not the whole probability question. D-VBLL at $1/T$ ties MAP on accuracy, NLL, and error ranking. The ECE edge over unscaled softmax disappears once the softmax has a validation temperature. I should not describe D-VBLL as a better calibrated classifier than temperature scaling.
- Posterior variance can collapse. The prior said 0.016; training at $1/T$ produced about 5×10^{-4} . After that, sampling the last layer cannot invent distance-aware uncertainty. Raising the KL weight reverses the collapse: entropy and MNIST AUROC go up, clean ECE gets worse, and accuracy on this CNN barely moves.
- A Bayes classifier still needs a representation. Pixel naive Bayes scores 0.579 because garment pixels are not independent Gaussians. LDA on the CNN features jumps to 0.917, and the discriminative heads then cut NLL from 0.544 to 0.200. The modern part is the feature map; the last-layer posterior is a small adjustment on top.
- Confidence is not a corruption alarm, but it is a reject option on clean data. Low contrast keeps the garment shape, both models stay confident, and ECE exceeds 0.65. On clean test images, dropping the least confident 10% lifts accuracy to about 0.967 for both heads. Shirt versus T-shirt is a label-overlap problem, not a Bayes problem.
- Out-of-distribution AUROC needs more than one seed. MAP’s two seeds differed by about nine points. A single-run bar chart would have picked a winner that the second seed does not support.

6 Source Code Webpage Link

Source code:

<https://github.com/AngusJai/CISC3024-AI-Assignment2>

The repository contains `models/`, `train.py`, `evaluate.py`, `run_all.py`, `run_ablation.py`, `process_log.md`, `report.tex`, and the metrics and figures under `outputs/`. Dataset downloads (`data/`) and virtual environments are gitignored. Trained checkpoints for the main seeds are included so `demo.py` can be run without

retraining.

7 Conclusion

D-VBLL is a recent Bayesian classifier that an agent can implement from a variational bound and test on clothing images in one afternoon. With the paper’s $1/T$ KL weight it reproduces the softmax baseline’s accuracy and slightly improves unscaled calibration. A validation temperature on that softmax calibrates further, and a classical LDA head on the same features is close in accuracy but worse in NLL. D-VBLL does not automatically detect digit images or low-contrast clothes, because the approximate posterior concentrates and the predictive distribution collapses toward the mean network. The KL ablation makes that statement empirical: turn the KL up and entropy and digit-detection AUROC rise, while clean calibration falls. Accuracy on this network stays near 0.93.

References

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